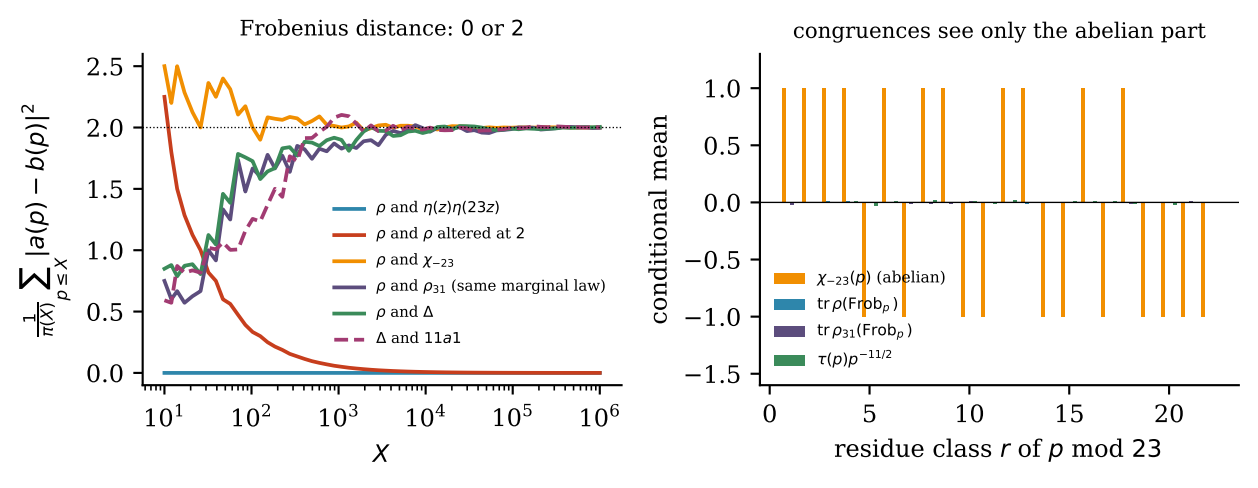
Frobenius distance: 0 or 2
$\frac{1}{\pi(X)} \sum_{p \le X} |a(p) - b(p)|^2$
2.5
2.0
1.5
1.0
0.5
0.0
10^1
10^2
10^3
10^4
10^5
10^6
X
ρ and $\eta(z)\eta(23z)$
ρ and ρ altered at 2
ρ and χ_{-23}
ρ and ρ_{31} (same marginal law)
ρ and Δ
Δ and 11a1
congruences see only the abelian part
conditional mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
5
10
15
20
residue class r of p mod 23
$\chi_{-23}(p)$ (abelian)
tr ρ(Frob$_p$)
tr ρ_{31}(Frob$_p$)
$\tau(p)p^{-11/2}$